

Architecture Selection – Community Tool & Equipment Library

Architectural Choice

We chose **Layered Architecture** for the Community Tool & Equipment Library System. The system is organized into Presentation, Business, and Data layers. The Presentation Layer handles member interaction, the Business Layer manages orders, equipment and payment operations, and the Data Layer manages persistent system information.

Reason 1 – Separation of Concerns

Layered Architecture separates user interaction from business operations and data management. This makes functions such as equipment search, reservation, checkout, payment processing, and return tracking easier to organize.

Reason 2 – Maintainability

Each layer has a specific responsibility. For example, changes to the user interface can be made without directly changing the database component, making the system easier to maintain.

Security Advantage

The layered structure allows access to member, reservation, payment, and equipment information to be controlled through the business layer rather than providing unrestricted direct access to stored data.

Performance Benefit

The separation of components allows frequently used operations such as equipment availability checks and database operations to be handled by dedicated components, keeping communication between system parts organized.